

U-Net for Image Reconstruction (Denoising)

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Abstract

This research evaluates the U-Net deep learning model for reconstructing galaxy images from the DECals dataset, subjected to Gaussian noise (standard deviation $\sigma = 10$, mean zero), comparing it to traditional techniques like DCT and DWT enhanced by the Iterative Shrinkage-Thresholding Algorithm (ISTA). The goal is to identify which method best reconstructs galaxy images under added noise. Initial results indicate U-Net outperforms other methods with this specific dataset, although its effectiveness may vary with different datasets. This highlights U-Net's variable performance depending on data characteristics.

Introduction

This study assesses the U-Net convolutional neural network's effectiveness in reconstructing detailed galaxy images from the DECals dataset, subjected to Gaussian noise, compared to traditional methods like DCT and DWT enhanced by the Iterative Shrinkage-Thresholding Algorithm (ISTA). We explore whether U-Net's advanced image processing capabilities outperform these methods, which typically reduce image size and noise but may obscure critical details.

We present the mathematical foundations of each method, their processing approaches, and experimental results. The findings highlight performance differences and the potential of using

advanced deep learning for astronomical image reconstruction, offering insights into optimal techniques and suggesting directions for future research.

U-Net Architecture Overview (Mathematical representation)

Contracting Path (Encoder)

Convolutional Layers: For an input image denoted by I , the convolution operation is described as follows:

$$F_k = \sigma(W_k * I + b_k)$$

Here, F_k represents the feature map produced by applying the k -th filter, W_k is the weight matrix for the k -th filter, $*$ indicates the convolution operation, b_k is the bias, and σ is a nonlinear activation function such as ReLU.

Expansive Path (Decoder)

Up-Convolution (Transposed Convolution): This step increases the resolution of the output:

$$U = \sigma(W^T * F + b)$$

Where U is the up sampled feature map, W^T represents the transposed weight matrix, and F is the input feature map.

Convolutional Layers: Additional convolutional layers refine the upscaled feature maps, integrating local context to improve the output quality:

$$R = \sigma(W * (U \oplus C) + b)$$

Here, R is the final refined output, U is the up sampled feature map, C is the concatenated feature map from earlier layers, and $\oplus$ denotes the concatenation operation.

Final Layer:

Output Layer: The network typically concludes with a 1x1 convolution that maps each vector of the feature map to the desired output, such as different classes for segmentation or specific values for reconstruction:

$$O = \sigma(W * R + b)$$

O represents the final output, be it an image or a segmentation map.

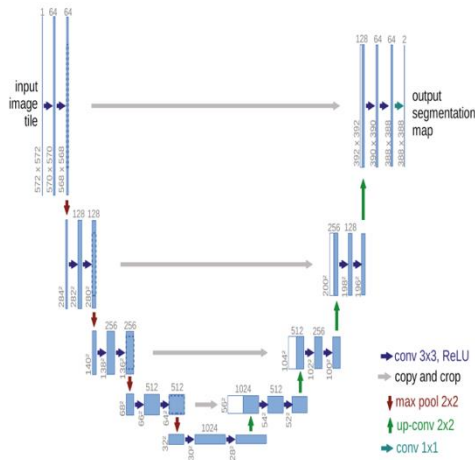


Fig. 1. U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

Experiments and Results

Experimental Setup and Rationale

In order to mimic real-world imaging issues, this study uses a U-Net convolutional neural network to rebuild galaxy images from the DECals dataset that have been digitally manipulated with Gaussian noise. The architecture of U-Net is perfect for analyzing complicated astronomical images because it is built for accurate feature extraction.

With layers employing ReLU activations and dropout to avoid overfitting, the network has expanding and contracting routes that culminate in a sigmoid output layer. In order to evaluate the model's performance with different levels of data deterioration, we test it under two different conditions: 2% and 5% pixel retention.

To preserve feature integrity, images are preprocessed to 256 by 256 pixels, normalized, and only slightly enhanced. The training is guided by the mean squared error loss and Adam optimizer, which balance the dangers of overfitting with efficient learning. capacity to tolerate varying degrees of image quality deterioration, proving both its best performance and usefulness in difficult circumstances.

Methodology

In order to meet the input specifications of the U-Net model, our experimental methodology preprocesses the galaxy images by downsizing them to 256x256 pixels and normalizing the pixel values. The model's capacity to generalize from noisy input is then improved by using a data augmentation technique. This includes simulated changes, however they are maintained to a minimum to preserve the authenticity of the astronomical features, such as small rotations and shifts.

The Adam optimizer and mean squared error loss are used to train the U-Net model, making it appropriate for regression applications such as image reconstruction. A moderate learning rate and a restricted number of training epochs were chosen as training parameters to balance training efficiency with model performance and reduce the risk of overfitting.

Presentation of Results with Comparative Images

This section presents the experimental results and illustrates the superiority of U-Net over more conventional techniques, such as Discrete Cosine Transform (DCT) and Discrete Wavelet Transform (DWT) enhanced with Iterative Shrinkage-Thresholding Algorithm (ISTA), for denoising galaxy images.

Comparative Image Panels: We choose a representative set of galaxy images from the DECals dataset that have undergone artificial noise addition. Each image set includes:

1. The original image with added noise.
2. The image denoised using DCT with ISTA.
3. The image denoised using DWT with ISTA.

4. The image denoised using U-Net.

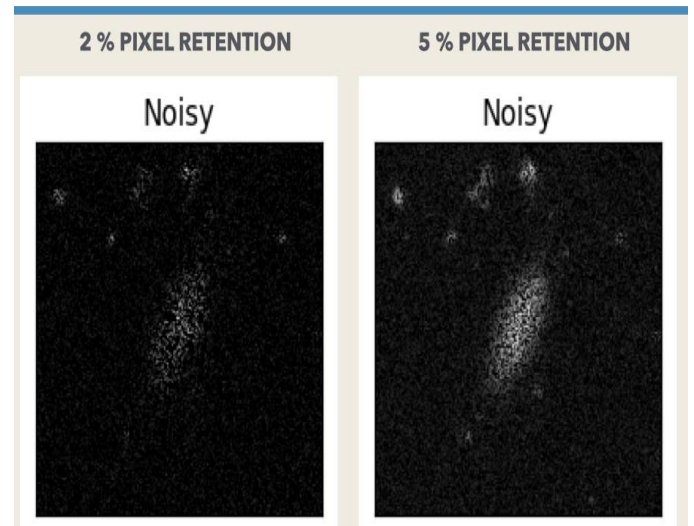


Figure 1 The original image with added noise

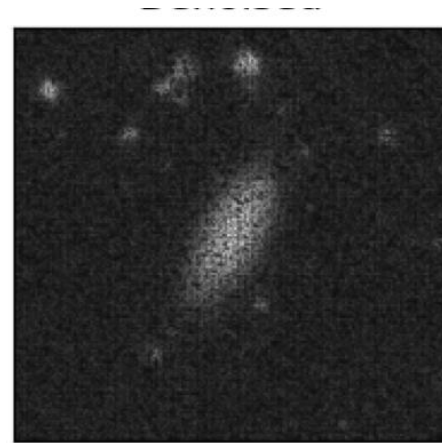


Figure 2 The image denoised using U-Net (5% Pixel)

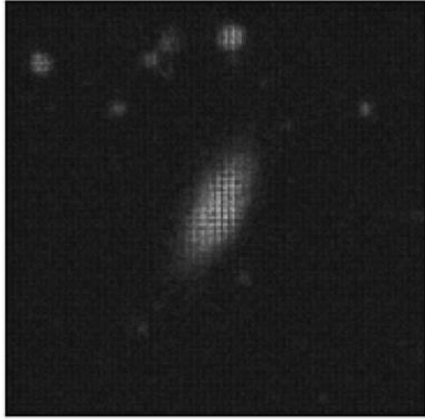


Figure 3 The image denoised using U-Net(2% Pixel)



Figure 4 The image denoised using DWT with ISTA

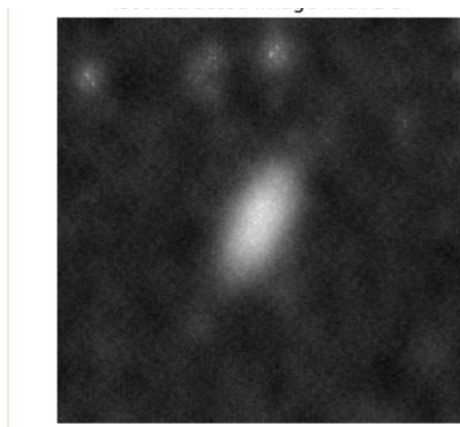


Figure 5 The image denoised using DCT with ISTA

Conclusion from Experiments

This work shows that when it comes to reconstructing galaxy images from the DECals dataset, U-Net performs noticeably better than conventional denoising techniques like DCT and DWT augmented by ISTA. Important conclusions consist of:

Performance Overview:

- **Reconstruction Accuracy:** U-Net delivers superior accuracy, producing higher quality reconstructions of galaxy images.
- **Detail Preservation:** U-Net maintains important image details even in the presence of high Gaussian noise, contrasting with the over smoothing tendency of traditional methods.

Comparative Analysis:

- **Visual Quality:** Visual comparisons indicate that U-Net reconstructs clearer and more detailed images, showing its capability to handle complex structures better than DCT and DWT+ISTA.

Insights & Implications:

- **Versatility and Impact:** U-Net's robustness against various noise levels suggests potential for extensive applications in noisy astronomical data processing, which could significantly enhance astrophysical and cosmological research.

Future Directions:

- Real-time Processing: Exploring U-Net's integration into real-time systems could improve live astronomical data analysis.
- Expanded Training: Further training on larger, diverse datasets may help U-Net generalize better across different astronomical scenarios, boosting its effectiveness.

In summary, U-Net's advanced performance in image detail preservation and adaptation to noise levels makes it a superior tool for enhancing the accuracy and depth of astronomical analyses.

References

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